Part 3: Introduction to Modelling

A fundamental step in many data analyses is to model raw, observable data, i.e. to replace the complex processes that generate these data with a mathematically and computationally simpler fiction.

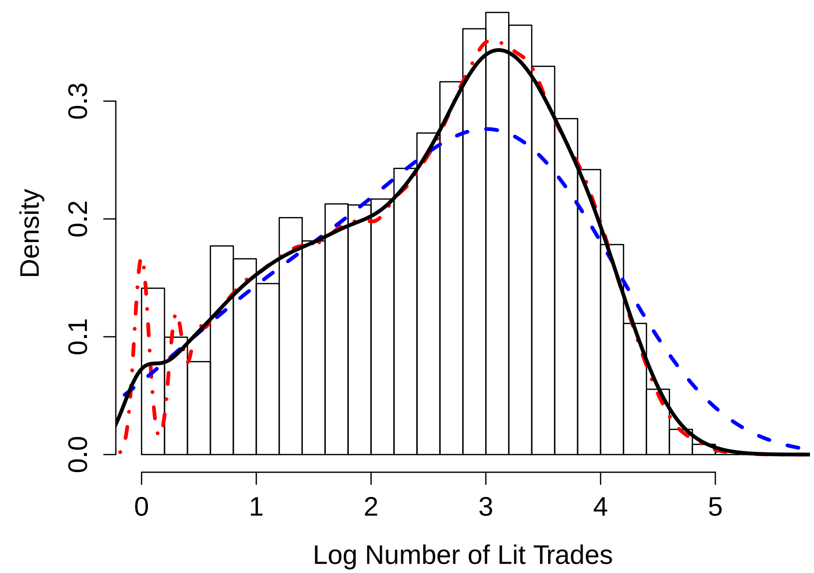
You have likely heard the expression, "All models are wrong; some are useful."[1] That is a good mentality to take towards the modelling process: We do not believe that the data were actually generated by, for example, drawing from a normal distribution. Instead, it **may** be the case that the normal provides a sufficiently accurate approximation to that process.

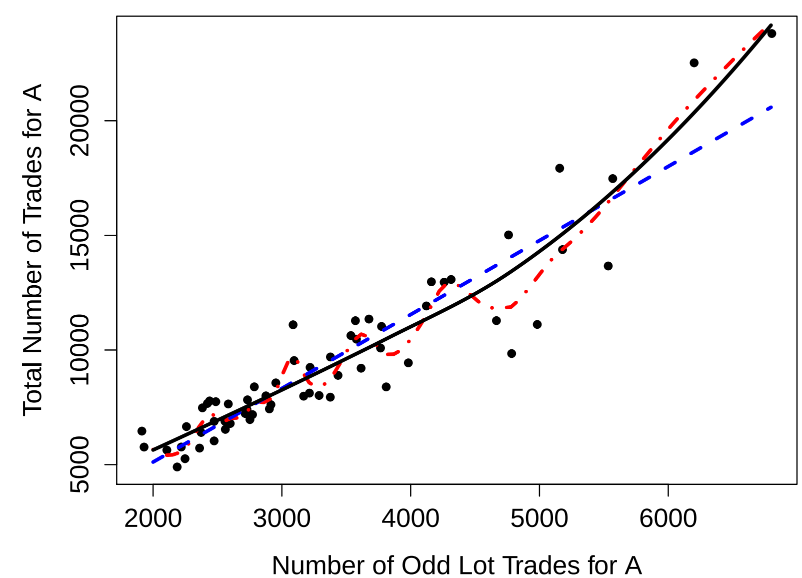
[1]For a history of this saying, see https://en.wikipedia.org/wiki/All_models_are_wrong.

A key heuristic in the statistical modelling process is that of smoothness.

We believe that most relationships, and most distributions, are fundamentally smooth, i.e., we don't try to fit to every little fluctuation that our sample may possess.

Of course, this idea can be taken too far, and oversmoothing can lead to missing important features that are present in data.





Exercise: Explain how the three proposed models on the previous slides seem to perform, in regards to the appropriate amount of smoothing of the histogram and scatter plot.

Exercise: Explain why overfitting is such a significant concern.

In many situations, there are one or more unknown population quantities that are of interest to estimate. For example,

- The probability that the return exceeds c tomorrow
- The "beta" for some stock over the past year.
- The volatility for some stock over the past three months.
- The mean of some distribution of interest.
- The variance of some distribution of interest.
- The λ parameter from some Exponential distribution.

The model serves as a means to estimating such a quantity, since we imagine that the true model, if known, would provide the true value of the unknown. An estimated model provides an estimate of the unknown.

From a statistical standpoint, we say that a probability model is incompletely specified. I.e., a particular model is assumed, but there are still unknown components, often real-valued parameters.

We use available data (a sample) to estimate these parameters.

The quantities of interest such as those listed above are either the parameters themselves, or some function of those parameters.

Example: In a certain situation, an individual is comfortable assuming that $X_1, X_2, \ldots X_n$ are independent and identically distributed (i.i.d.) with the Normal distribution with mean μ and variance σ^2. These parameters are unknown, and must be estimated from data.

Exercise: Suppose someone wants to estimate the probability that tomorrow's return will be less than c. Describe steps as to how a model could be constructed and the quantity of interest estimated.

Terminology

The parameters are the constants that distinguish probability models in the same "family." Examples of "families" include "normal," "exponential," and so forth.

A statistic is a function of the RVs being used to model the observable data. Examples include the sample mean, the sample variance, etc. A statistic is a **random variable**.

An estimator is a statistic which is used to estimate an unknown parameter or an unknown function of a parameter. It is also a **random variable**.

The "hat notation" is standard for denoting an estimator: $\widehat{\beta}$ is an estimator for β, for example. θ is often used to denote a generic parameter, $\widehat{\theta}$ is its estimator.

Exercise: Explain the sense in which an estimator is appropriately thought of as a random variable.

Performance of Estimators

Our objective is to estimate unknowns, i.e., to construct estimators. For any situation, there are many choices of estimator, so we require methods to assess the performance of an estimator.

The "performance" is closely tied to the distribution of the estimator. The distributions we learned in Probability (plus some more) are useful for this purpose.

An ideal estimator $\widehat{\theta}$ for θ would have two properties:

1. The estimator should be accurate, i.e., its distribution would be centered on the true value θ.

2. The estimator should be precise, i.e., its distribution has a small spread (small variance)

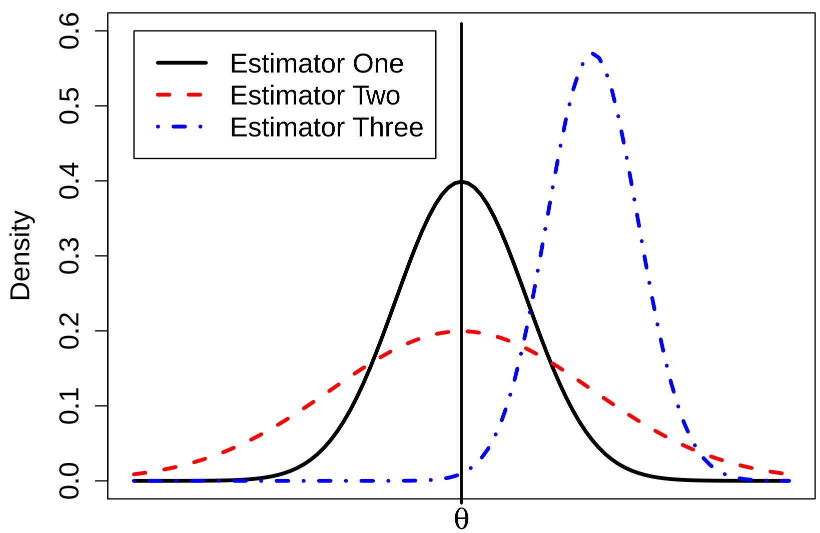
Example: The figure on the following page compares the distributions of three hypothetical estimators for a parameter θ, call them $\hat{\theta}_1$, $\hat{\theta}_2$, and $\hat{\theta}_3$.

We note that $E(\hat{\theta}_1) = \theta$, so it is optimally accurate. Also, $E(\hat{\theta}_2) = \theta$.

But, $V(\hat{\theta}_1) < V(\hat{\theta}_2)$, so $\hat{\theta}_1$ is more precise than $\hat{\theta}_2$.

So, we would prefer $\hat{\theta}_1$ to $\hat{\theta}_2$.

Also, $V(\hat{\theta}_3) < V(\hat{\theta}_1)$, but $\hat{\theta}_3$ has terrible accuracy.



An estimator $\hat{\theta}$ is said to be an unbiased estimator for θ if $E(\hat{\theta}) = \theta$.

The variability in an estimator (its precision) is naturally quantified by its variance, $V(\hat{\theta})$.

The standard error (SE) of $\hat{\theta}$ is the square root of its variance. The SE is often reported along with an estimate in order to give a sense of the amount of error one should anticipate in that estimate.

An estimator $\hat{\theta}$ is said to be the minimum variance unbiased estimator for θ (MVUE) if $\hat{\theta}$ is unbiased for θ and for any other unbiased estimator $\hat{\theta}'$, $V(\hat{\theta}) \leq V(\hat{\theta}')$.

Exercise: Suppose that X has the binomial(n, p) distribution. It is common to estimate p using $\hat{p} = X/n$, the sample proportion. Is this estimator unbiased? What is its variance and standard error?

There is often a "tradeoff" between bias and variance (accuracy and precision). This is called the bias-variance tradeoff.

In some cases, allowing some bias leads to reduced variance.

This tradeoff is made formal via the mean squared error (MSE). For an estimator $\hat{\theta}$ for θ,

$$\mathrm{MSE}_\theta(\hat{\theta}) = E\left((\hat{\theta}-\theta)^2\right)$$

i.e., it is the expected squared error in estimating θ when using $\hat{\theta}$.

One can show that

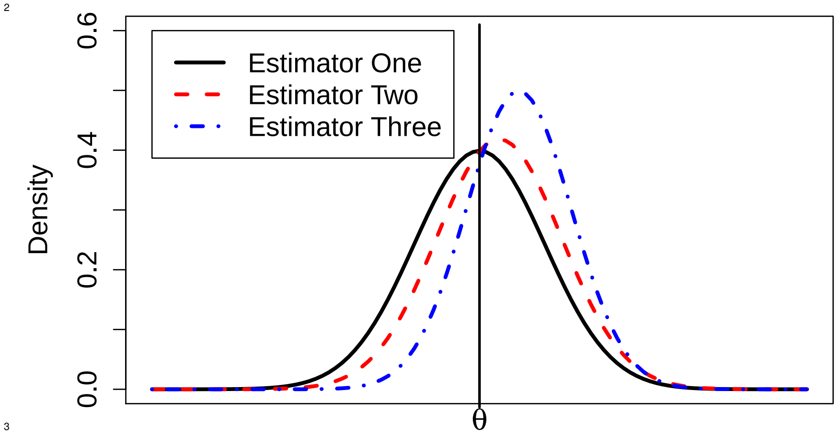
$$\mathrm{MSE}_\theta(\hat{\theta}) = V(\hat{\theta}) + \mathrm{bias}^2_\theta(\hat{\theta})$$

where

$$\mathrm{bias}_\theta(\hat{\theta}) = E(\hat{\theta}) - \theta$$

Exercise: How does the bias/variance tradeoff relate to our initial discussion of smoothness?

In the example below, all three estimators have the same MSE.



Important Cases

There are a few estimation situations that worth special attention.

Case 1: The first we have already explored, that of estimating a population proportion. We found that the sample proportion is unbiased and has a standard error of

$$\sqrt{\frac{p(1-p)}{n}}$$

Of course, we do not know the value of p, so typically it is replaced by its best estimate and the approximate SE is reported as

$$\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Exercise: We have derived the mean and variance of the sample proportion, but what is its distribution?

Case 2: Suppose that $X_1, X_2, \ldots, X_n$ are random variables and $E(X_i) = \mu$ and $V(X_i) = \sigma^2$ for all i. Further assume that these random variables are uncorrelated (meaning that every pair of random variables is uncorrelated).

Then, the standard estimator for μ is the sample mean, $\overline{X}$, and the standard estimator for σ^2 is the sample variance,

$$s^2 = \sum_{i=1}^{n} \left(X_i - \overline{X}\right)^2 \Big/ (n-1)$$

Note that the case where $X_1, X_2, \ldots, X_n$ are i.i.d. is a special instance of this situation.

Exercise: Derive the mean, variance, and standard error of the sample mean. What is its MSE as an estimator of μ?

Exercise: If we did assume that $X_1, X_2, \ldots, X_n$ are i.i.d., what could we say about the distribution of the sample mean?

Exercise: As defined, the sample variance is an unbiased estimator of σ^2. (The proof is not difficult, but a little tedious.)

Consider the following question: What choice of a minimizes the quantity

$$\sum_{i=1}^{n} (X_i - a)^2 \ ?$$

How does this result suggest it would be a good idea to estimate σ^2 by dividing the sum of squares by a constant less than n?

Case 3: Suppose that $X_1, X_2, \ldots, X_n$ are i.i.d. with the normal distribution with mean μ and variance σ^2.

Of course, this is a special instance of "Case 2" presented above, and the sample mean and sample variance remain unbiased estimators of μ and σ^2, respectively.

We can make some stronger statements, as well.

First, we know that $\overline{X}$ has **exactly** the normal distribution with mean μ and variance σ^2/n for **any** sample size.

Further results depend on properties of probability distributions that we will present now.

The Chi-squared distribution with k degrees of freedom is a special case of the Gamma distribution. In particular, if X has the Gamma$(k/2, 1/2)$ distribution, then X has the Chi-square distribution with k degrees of freedom.

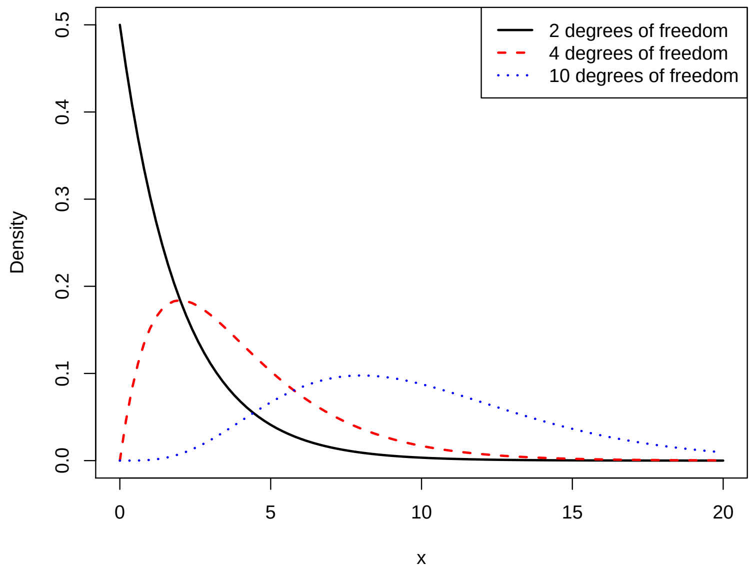
Also, if $Z_1, Z_2, \ldots, Z_k$ are i.i.d. standard normal random variables, then

$$X = \sum_{i=1}^{k} Z_i^2$$

has the Chi-squared distribution with k degrees of freedom.

If X has this distribution, then $E(X) = k$ and $V(X) = 2k$.

Plots of pdfs from the Chi-squared distribution are shown on the following slide.



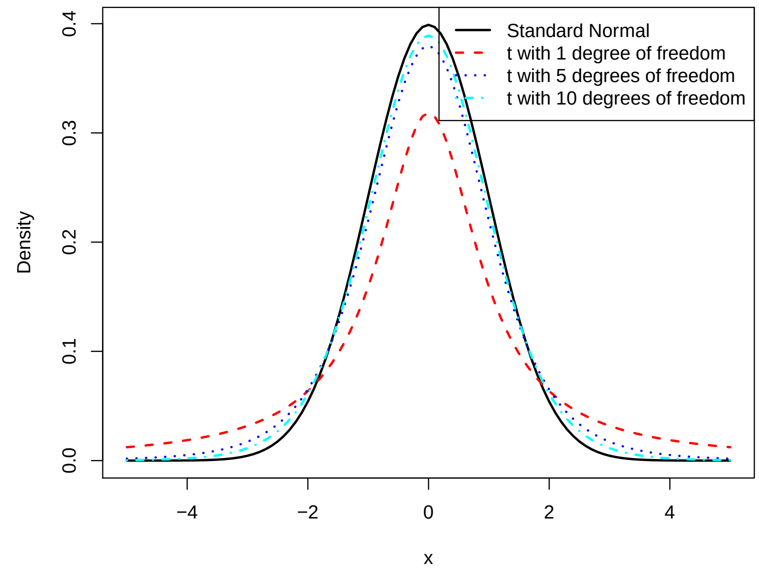
The t-distribution with ν degrees of freedom is defined to be a random variable T that can be constructed as

$$T = \frac{Z}{\sqrt{X/\nu}}$$

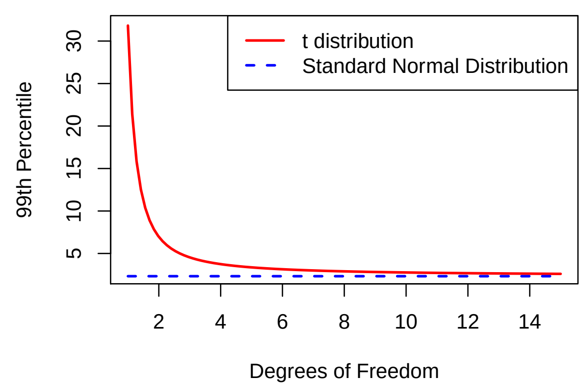
where Z is a random variable with the standard normal distribution, and X is a random variable with the Chi-squared distribution with ν degrees of freedom. Further, Z and X are assumed independent.

This distribution has a "normal-shaped" pdf, but has heavier tails than the normal distribution. As ν increases, the distribution converges to the standard normal distribution.

It holds that $E(T) = 0$ when $\nu \geq 2$ and that $V(T) = \nu/(\nu - 2)$ when $\nu \geq 3$.



The figure below makes clearer how much additional probability is in the tails of the t-distribution compared to the standard normal. For $\nu = 1$, 1% of the observations will be larger than 30. The 99th percentile of the standard normal distribution is 2.33.



Now we can state the four **classic results** that hold under our "Case 3:"

Assuming that $X_1, X_2, \ldots, X_n$ are i.i.d. Normal(μ, σ^2), then

1. $\overline{X}$ and s^2 are independent.
2. The quantity
$$\frac{\overline{X} - \mu}{\sigma/\sqrt{n}}$$
has the standard normal distribution.
3. The quantity
$$\frac{(n-1)s^2}{\sigma^2}$$
has the Chi-squared distribution with $n-1$ degrees of freedom.
4. The quantity
$$\frac{\overline{X} - \mu}{s/\sqrt{n}}$$
has the t-distribution with $n-1$ degrees of freedom.

Exercise: Under "Case 3" it is easier to show that s^2 is an unbiased estimator of σ^2, and its also easy to derive the variance of this estimator. Do these now.

Confidence Intervals

The already-mentioned standard error (SE) is a useful tool for quantifying error in an estimate, but a common alternative is the use of confidence intervals for unknown parameters.

Formally, we call (L, U) a $100(1-\alpha)\%$ confidence interval for θ if
$$P(L \leq \theta \leq U) = 1 - \alpha$$

Hence, if want a "95% confidence interval," set $\alpha = 0.05$. Other common choices are 90% and 99% confidence intervals (α = 0.10 and $\alpha = 0.01$, respectively).

An appropriate statement regarding a confidence interval is, for example, "I am 95% confident that the true value of θ is between 10.6 and 11.7."

It is **not** appropriate to say that "The probability that θ is between 10.6 and 11.7 is 0.95." This is incorrect because there is nothing random in that statement: θ, 10.6 and 11.7 are all constants.

The randomness is in the procedure itself. If one constructs 95% confidence intervals in a wide range of situations, then 95% of those intervals will cover the true value of the parameter.

There is an alternative approach to statistical inference, Bayesian inference, in which parameters are treated as random variables. More on that later.

Exercise: Use our “classic results” above for “Case 3” to derive confidence intervals for μ and σ^2 in that situation. For estimating μ, onsider both the instances where σ^2 is known and unknown.

The next slides present four important confidence intervals that are commonly used.

Confidence Interval for Population Mean, μ, when n is large $(n \geq 30)$

Assume that the sample is i.i.d. from a distribution with mean μ and variance σ^2, with both μ and σ^2 unknown. Then, a $100(1-\alpha)\%$ confidence interval for μ is

$$\overline{X} \pm z_{\alpha/2}\left(\frac{s}{\sqrt{n}}\right)$$

where z_a is such that $P(Z > z_a) = a$ when Z has the standard normal distribution. In R notation, z_a equals `qnorm(1-a)`.

Confidence Interval for Population Mean, μ, when n is small $(n < 30)$

Assume that the sample is i.i.d. from the normal distribution with mean μ and variance σ^2, with both μ and σ^2 unknown. Then, a $100(1-\alpha)\%$ confidence interval for μ is

$$\overline{X} \pm t_{\alpha/2,n-1}\left(\frac{s}{\sqrt{n}}\right)$$

where $t_{a,v}$ is such that $P(T > t_{a,v}) = a$ when T has the t-distribution with v degrees of freedom. In R notation, $t_{a,v}$ equals `qt(1-a,v)`.

Confidence Interval for Population Variance, σ^2

Assume that the sample is i.i.d. from the normal distribution with mean μ and variance σ^2, with both μ and σ^2 unknown. Then, a $100(1-\alpha)\%$ confidence interval for σ^2 is

$$\left(\frac{(n-1)\,s^2}{\chi^2_{\alpha/2,n-1}}, \frac{(n-1)\,s^2}{\chi^2_{(1-\alpha/2),n-1}}\right)$$

where $\chi^2_{a,v}$ is such that $P(U > \chi^2_{a,v}) = a$ when U has the chi-squared distribution with v degrees of freedom. In R notation, $\chi^2_{a,v}$ equals `qchisq(1-a,v)`.

Confidence Interval for Population Proportion, p

Assume that n is large enough that you are confident that $np \geq 5$ and $n(1-p) \geq 5$. Let $\hat{p} = X/n$ where X is binomial(n,p), and p is unknown. Then, a $100(1-\alpha)\%$ confidence interval for p is

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

where z_a is such that $P(Z > z_a) = a$ when Z has the standard normal distribution. In R notation, z_a equals `qnorm(1-a)`.

Exercise: A sample of size 50 is drawn from a population with unknown mean and variance. It holds that $\overline{x} = 14.6$ and $s = 1.3$. Estimate the population mean. State both the standard error for the estimator and a 90% confidence interval for μ.